IMT Mines Alès
École Mines-Télécom

RÉPUBLIQUE FRANÇAISE
Liberté
Égalité
Fraternité

DevOps for Digital Twins and by Digital Twins: Leveraging Virtual Replicas in Continuous Software Engineering

GT Jumeaux Numériques – GDR SciLog

17 mars 2026

Sylvain Vauttier – IMT Mines Alès – Laboratoire SyCoIA

Institut Mines-Télécom
IMT Mines Albi École Mines-Télécom
IMT Atlantique Bretagne-Pays de la Loire École Mines-Télécom
IMT Nord Europe École Mines-Télécom IMT-Université de Lille
Institut Mines-Télécom Business School
MINES Saint-Étienne
TELECOM Paris
TELECOM SudParis
EURECOM
InSIC

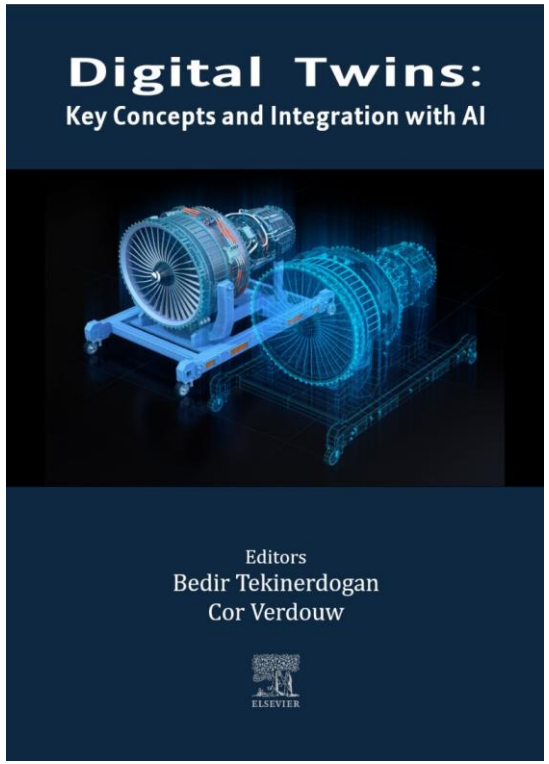
Context

Accepted chapter for the book **"Digital Twins: Core Principles, Systems Engineering, and AI Integration"**

- To be published by Elsevier in june 2026

- Collective work. Mainly from members of the GT DT
  - Bérenger Akodo, Jannik Laval, DISP, Université Lyon 2 et 1, Insa Lyon
  - Gwendal Beaumont, Antoine Beugnard, Savador Martinez, Lab-STICC, IMT Atlantique,
  - Souad Rabah Chaniour, Christelle Urtado, Sylvain Vauttier, SyCoIA, IMT Mines Alès

- A position paper
  - Digital twins: an all-around paradigm for the digital transition
    - Not only for managing systems (digital replicas of systems, DevOps for DTs)
    - But also, for managing systems' life-cycles (DTs in/for DevOps, digital replicas of DevOps processes)

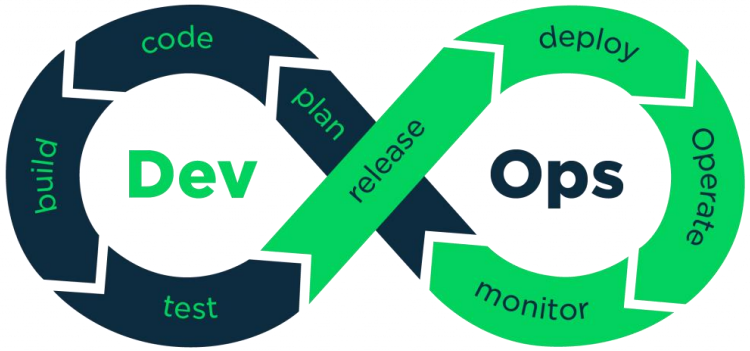


DevOps for DTs

Benefits of a generic DevOps approach

- Best practices for agile engineering of complex software systems

- Four pillars (Humble and Farley, 2010)
  - interdisciplinary collaboration: DTs mixes data, models, software and even IoT (CPS)
  - process automation: fast availabilty updates/corrections ; repeatable deployment of new versions
  - continuous monitoring: early detection of fidelity loss
  - iterative improvement: co-evolution of DTs and systems



Research questions

- What are the specific characteristics of DTs as software systems?
- How to adapt DevOps to support effectively the specific characteristics of DTs?

DevOps for DTs

What are the specific characteristics of DTs ?

- From an organic point of view
  - specific functions (monitoring, visualization, control, simulation…)
  - reference architectures
  - implementable with existing technologies (CPS, IoT, cloud computing, micro services…)

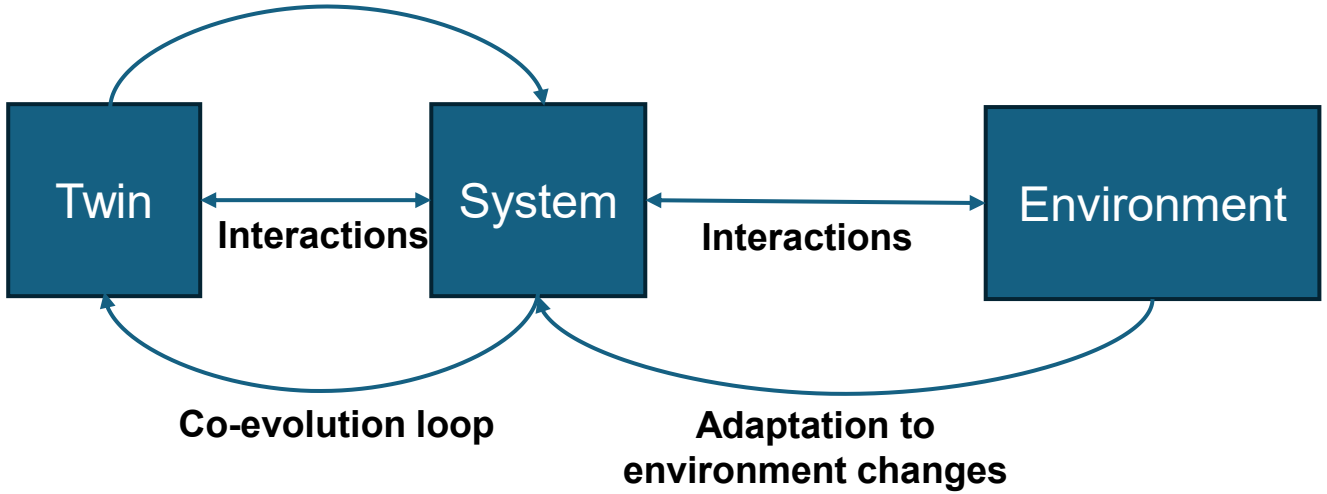
- From a conceptual point of view
  - DTs, as **digital replicas**, must be **synchronized with** their **reference systems**

Adapted DevOps for DTs

- Specific modeling tools, DSLs for developing and managing DTs
- Leveraging the DevOps automated process loop as a framework for monitoring and maintaining the coupling between a Digital Twin and its corresponding actual system

Three kinds of drifts (from litterature)

- State drifts
  - DTs must continuously keep up-to-date their representations of systems
- Model drifts
  - DTs must modify their models to adapt them to changes in systems' behaviors
- Architectural drifts
  - DTs must co-evolve and reflect the architectural changes of the systems to remain faithful digital replicas

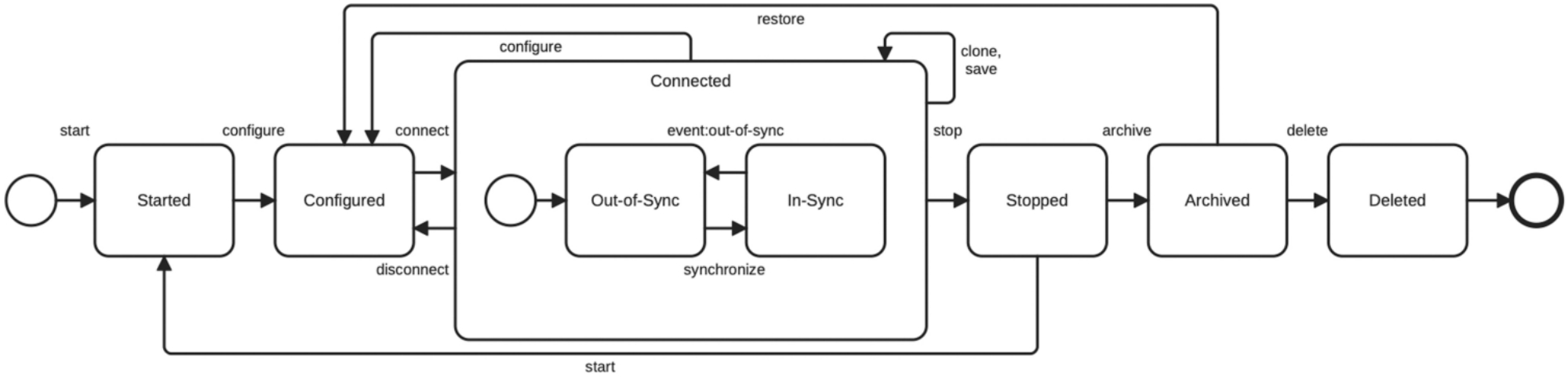


DevOps for DTs

On-going work

- Formalization of DTs' lifecycle management and mapping to DevOps frameworks
  - standardize procedures to facilitate the (re)-deployment and operation of DTs

- First focus: the "operate" part of DevOps processes
  - Formalization of DTs' lifecycle as a state machine

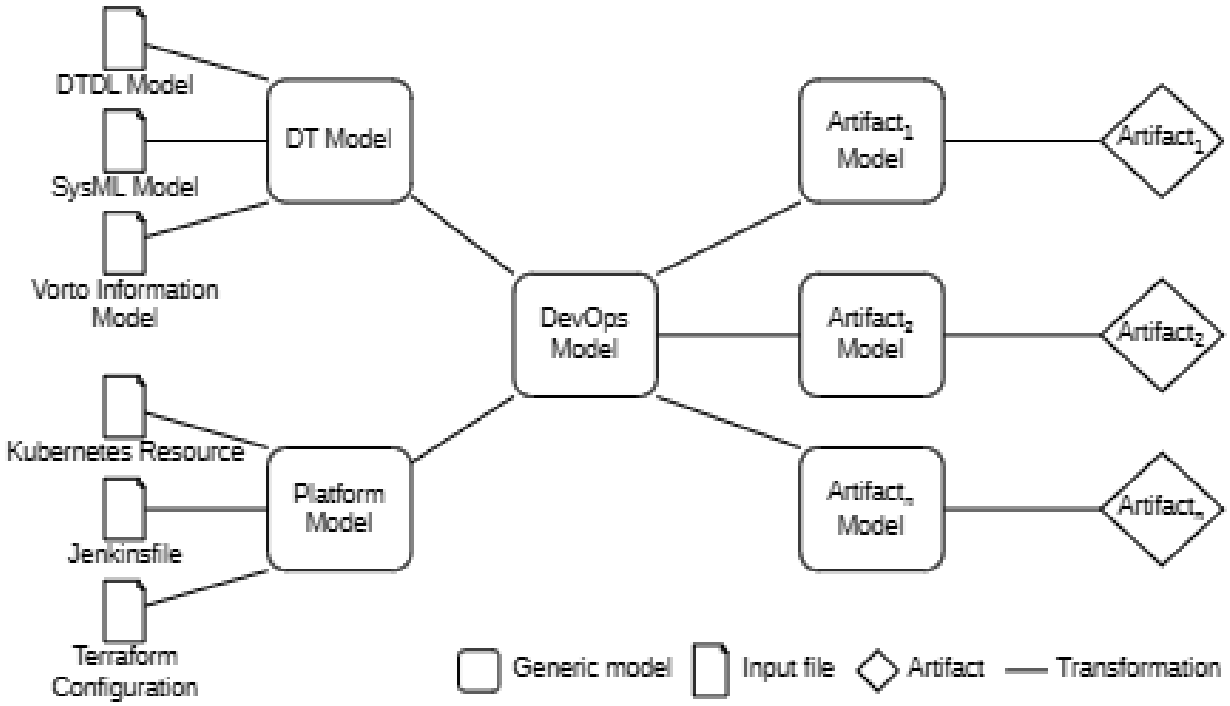


Gwendal Beaumont, Antoine Beugnard, Salvador Martínez, Christelle Urtado, Sylvain Vauttier. Towards Automating the Life Cycle Management of Digital Twins. ER2025 - 44th International Conference on Conceptual Modeling, Oct 2025, Poitiers, France. pp.412-430

DevOps for DTs

On-going work

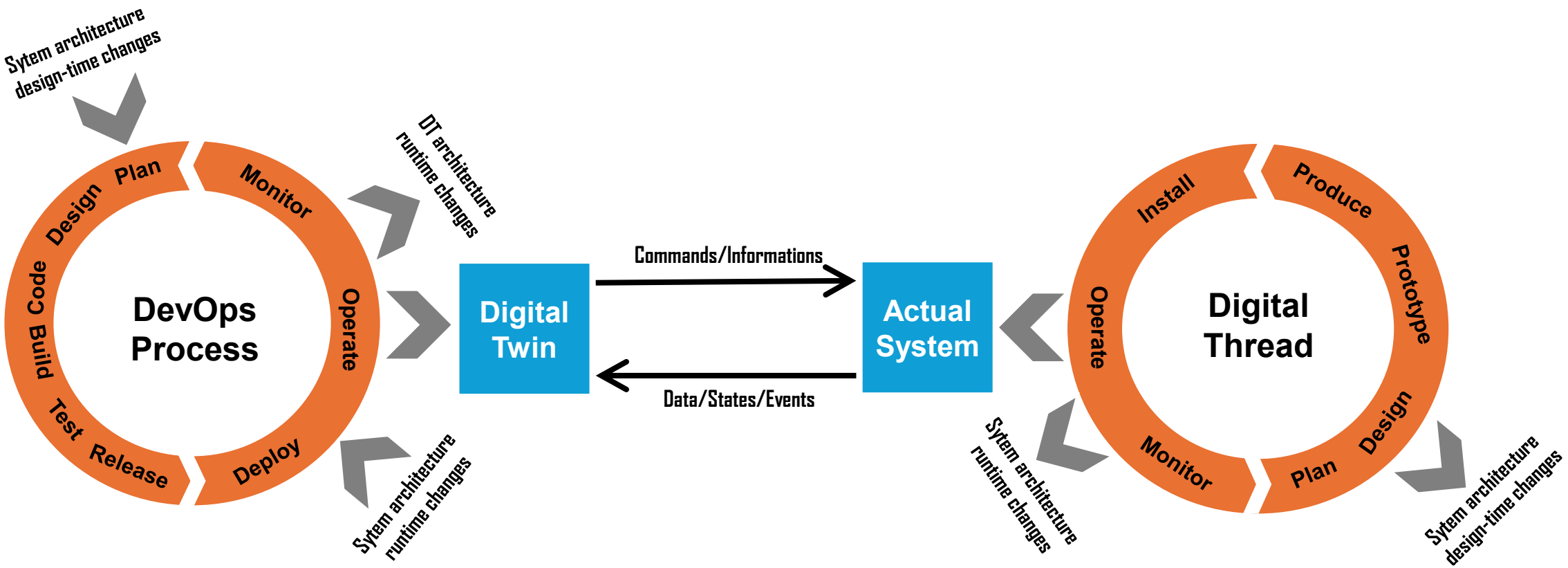
- First focus: the “operate” part of DevOps processes
  - Implemented as a model-driven engineering framework based on pivot models and transformations
  - Experimented on a FischerTechnik SLD module, with DTDL and GitOps



Gwendal Beaumont, Antoine Beugnard, Salvador Martínez, Christelle Urtado, Sylvain Vauttier. Towards Automating the Life Cycle Management of Digital Twins. ER2025 - 44th International Conference on Conceptual Modeling, Oct 2025, Poitiers, France. pp.412-430

Future work

- Support for whole DevOps processes
- Synchronization/Integration of the DevOps processes of DTs with the Digital Threads of systems



DTs for DevOps

DevOps and DTs : a natural match…

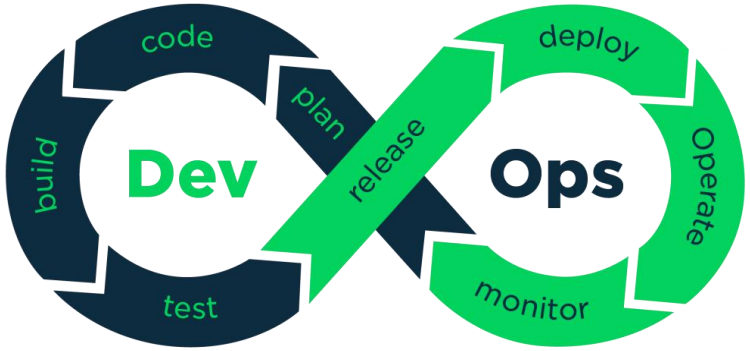
- DevOps relies on digitalizing software lifecycle management
  - feedback loop through monitoring of operations
  - continuous evolution / maintenance
- DT is the key concept for “virtual spaces” (Grieves, 2017)
  - virtual replicas for monitoring, visualization, control, simulation…

… being explored by the SE community

- continuously updated virtual representation of the target system (Marchand *et al.*, 2025a)
  - offline test and validation, but on realistic and actualized conditions, in synch with the actual system
  - DevTestOps (Angara, Gutta and Prasad, 2018): continuous testing in the DevOps loop

Exemples of Potential Roles for DTs in DevOps phases

- Plan
  - requirement engineering based on dynamic visualization, analysis, simulation of systems
- Code/Build/Test
  - sandbox for developing and testing code using simulations (mocks) of target systems
  - evaluation of different configurations, operation scenarios
  - simulation of attacks, security breaches
- Release / Deploy
  - configuration management, evolution impact simulation and audit, using accurate representations of target systems
- Operate/Monitor
  - analysis and optimization of system performance by simulating different configurations and load increases
  - visualization and analysis of system status
  - (predictive) identification of issues

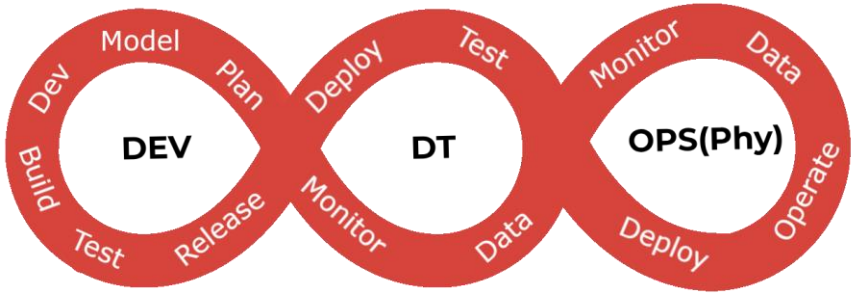


DTs for DevOps

MBSE for DTs in DevOps

- TwinOps (Hugues *et al.*, 2020): models as central artefacts for CPS engineering using DTs
  - Modeling phase: functional and non-function requirements definition, architectural design, virtual integration (interface design, verification and validation)
  - Testbench / system realization: generation of code for the simulation, the implementation and the instrumentation (monitoring probes) of the system; building of software for multiple targets (system, simulations)
  - Ops: deployment of the generated code ; collection of monitoring data ; analysis of the data

- DevTwinOps
  - Explicit integration of DTs in the DevOps process as a preproduction qualification stage



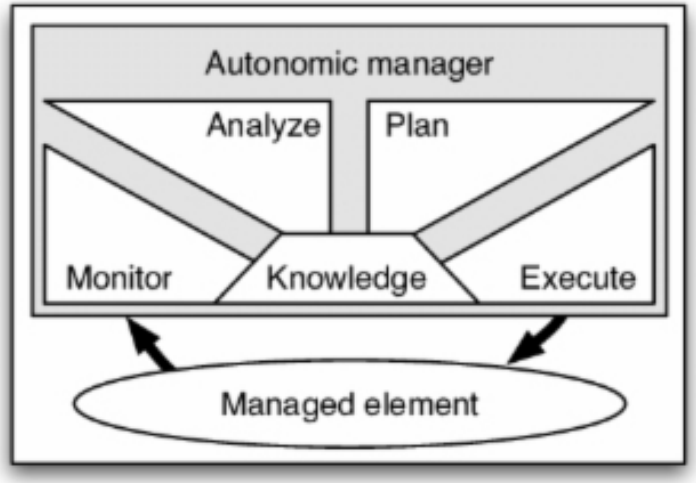
Conclusion

Towards reflexive or meta-DTs?

- DTs help engineering and managing systems

- DTs help engineering and managing the software of systems
  - support the relation between Dev and Ops in a continuous improvement loop

- What helps engineering and managing DTs as software systems?
  - MBSE and code generation for Dev
  - What for feedback from Ops?
    - DT of DT
    - DT of the DevOps process (activities, artefacts, …)



MAPE-K architecture for autonomous adaptation

IBM (2006) “An architectural blueprint for autonomic computing. IBM Corporation.

IMT Mines Alès
École Mines-Télécom

RÉPUBLIQUE FRANÇAISE
Liberté
Égalité
Fraternité

DevOps for Digital Twins and by Digital Twins: Leveraging Virtual Replicas in Continuous Software Engineering

GT Jumeaux Numériques – GDR SciLog

17 mars 2026

Sylvain Vauttier – IMT Mines Alès – Laboratoire SyCoIA

Institut Mines-Télécom · IMT Mines Albi École Mines-Télécom · IMT Atlantique Bretagne-Pays de la Loire École Mines-Télécom · IMT Nord Europe École Mines-Télécom IMT-Université de Lille · Institut Mines-Télécom Business School · MINES Saint-Étienne · TELECOM Paris · TELECOM SudParis · EURECOM · InSIC